COMP 3920
Written and Practical Final Review Sheet

Final format:

- Written Portion
- Practical Portion, using Node.js and Draw.io
- 2 hour time limit (for both portions combined)
- Hand in your Written Portion SELECT queries
- Practical Portion begins when you submit your Written Portion
- Submit your code/screenshots

Written Portion is **Closed Book**.

<u>What is Allowed</u>

☑ You may use a single 8 ½" x 11" double sided sheet of paper as a cheat sheet.

<u>What is NOT Allowed</u>

☒ You may NOT consult any of your class notes and/or previous labs/projects.
☒ You may NOT consult the Internet.
☒ You may NOT consult your textbook.
☒ You may NOT share any portion of your solutions with others.
☒ You may NOT use anyone else's solutions as your own.
☒ You may NOT have someone else complete any portion of your exam.
☒ You may NOT discuss your solution with others.
☒ You may NOT work with anyone else to complete the exam.

Practical Portion is **Open Book**.

- Connect to an AWS EC2 instance running MySQL
- Submit your screenshots

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What materials to review:

- Lesson Notes
- Lab Instructions
- Your Lab Reports
- Textbook Chapters
- Relationship Models and Implementation
- This Final Review Sheet

What do I need to know and be able to do?

- Interpret and create an ERD
    - Know what Entities, Relationships and Attributes are
    - How Primary Keys and Foreign Keys are represented
- Know and be able to follow the ERD and Database conventions:
    - Use singular in entity/table names
    - Use table name and _id for auto_increment/Identity PK columns
    - Name FKs with same name as PK in PK table
      Ex: person.person_id = pet.person_id
      (with a few exceptions for clarity and/or necessity)
    - Don't use . or spaces in entity/table names (use _ instead)
    - Always split M:M relationships into bridging table
    - Use lowercase letters for entity/table and attribute/column names
- Write SELECT statements to produce results that match sample screenshots
- Recognize examples of bad design. How would you fix them?
- Antipatterns
    - Jay-walking
    - Keyless entry
- Fix and write some simple Node.js code
    - Find errors in code and make the appropriate fix(es).
    - Augment the code to add additional functionality